

Assignment 1 – Mini Project

Detecting the anomalous activity of a ship's engine.

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Contents

Introduction.....	3
Approach.....	3
Analysis.....	5
Statistical Method	5
IQR Method	5
Machine Learning Methods	5
One Class Support Vector Machine (SVM).....	5
Isolation Forest	7
Results	8
Conclusion	8

Introduction

This mini project is to develop a robust anomaly detection system to protect a company's shipping fleet. Faults with the ship's engines can lead to malfunctions, downtime in the operations, potential safety hazards, breakdown in the ship's functionality all of which could have serious adverse consequences on the company's profitability by affecting revenue and customer satisfaction levels via failure to deliver goods on time.

The aim is to explore the data collected from the ships operations and employ statistical and feature engineering techniques to detect possible anomalies. Detecting anomalies will help the organisation to predict timely maintenance, thereby reducing downtime leading to increased profit, and increased customer satisfaction through timely deliveries.

Approach

The project used a combination of statistical and machine learning methods with the dataset provided with readings from the ship's operations, to detect anomalies.

The dataset included readings on the following features:

- Engine rpm (revolutions per minute)
- Lubrication oil pressure
- Fuel pressure
- Coolant pressure
- Lubrication oil temperature
- Coolant temperature

Initial exploration of the dataset indicated that there were no missing values or duplicated values within the dataset.

The table below shows the summary descriptive statistics for the dataset. The count shows that there are 19535 rows of data and the mean, median (50%), minimum and maximum values etc for each of the features within the dataset are also shown.

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp
count	19535.000000	19535.000000	19535.000000	19535.000000	19535.000000	19535.000000
mean	791.239263	3.303775	6.655615	2.335369	77.643420	78.427433
std	267.611193	1.021643	2.761021	1.036382	3.110984	6.206749
min	61.000000	0.003384	0.003187	0.002483	71.321974	61.673325
25%	593.000000	2.518815	4.916886	1.600466	75.725990	73.895421
50%	746.000000	3.162035	6.201720	2.166883	76.817350	78.346662
75%	934.000000	4.055272	7.744973	2.848840	78.071691	82.915411
95%	1324.000000	5.058040	12.208475	4.438415	84.940778	88.612891
max	2239.000000	7.265566	21.138326	7.478505	89.580796	195.527912

Table 1: Dataset Descriptive Statistics

The boxplots images of the six features are shown below and shows the outliers for each feature. The visual representation helps to see the outliers within the dataset. Identifying outliers can help to identify the presence of anomalies.

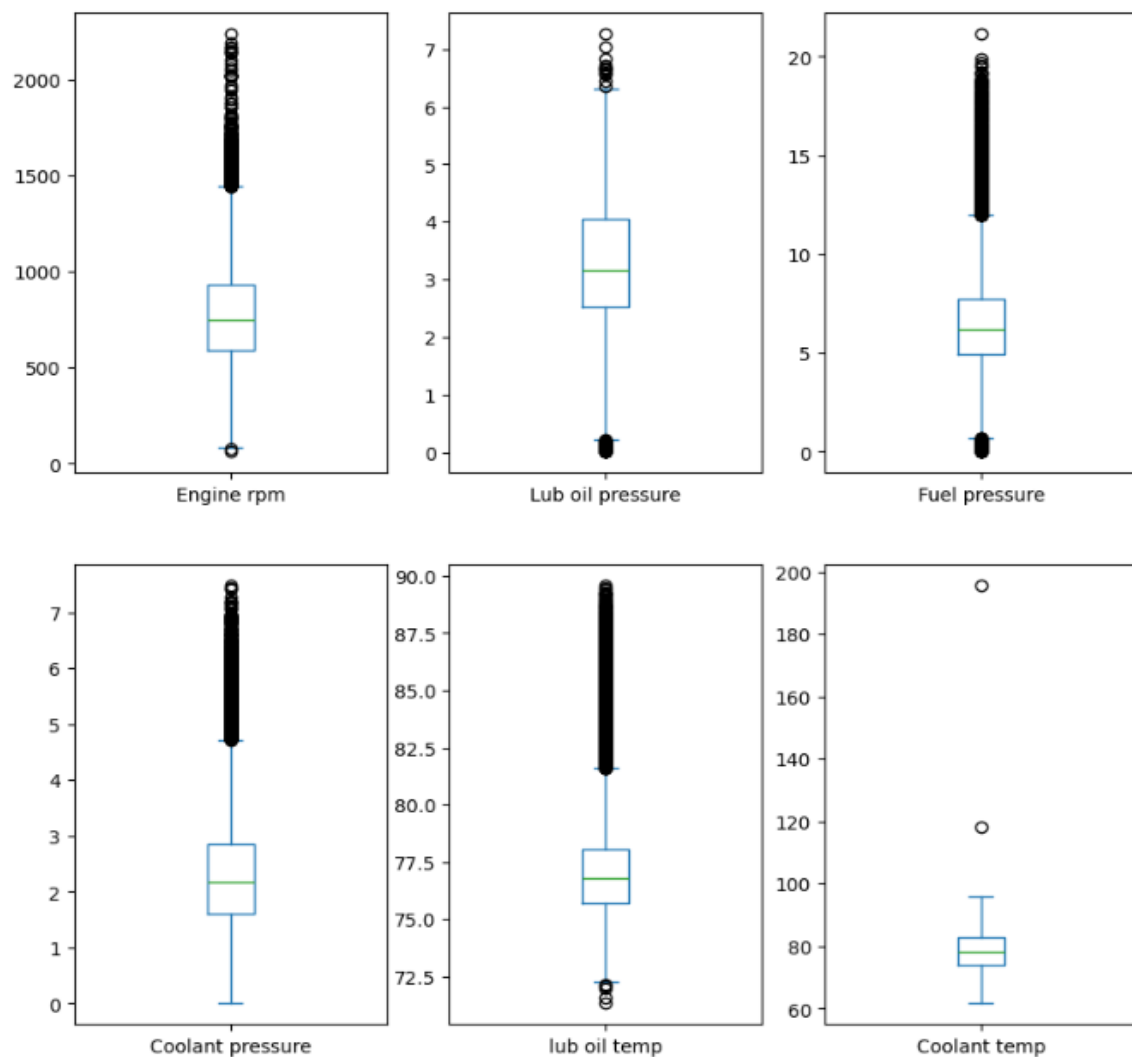


Figure 1: Boxplot visualisation for the dataset features.

Analysis

Statistical and machine learning methods were used for the anomaly detection.

Statistical Method

IQR Method

We calculated the IQR and values outside the IQR range for each feature are identified as outliers. The number of rows with outliers were identified. For this project, rows with a combination of more than 1 feature showing as outliers could be considered as anomalies.

The table below is a summary of the number of outliers identified using the IQR method

Number of features with outliers	Number of rows	Percentage of total data
Outliers in 1 or more feature	4636	23.73%
Outliers in 2 or more features	422	2.16%
Outliers in 3 or more features	11	0.05%
Outliers in 4 or more features	0	0

Table 2 : Outliers identified using IQR method

The data shows there were no rows within the data with 4 or more features as outliers. 11 rows had 3 feature outliers and all included outliers in three features - lubrication oil pressure, coolant pressure and fuel pressure.

Features with 2 or more outlier readings could be considered as anomalies which require further confirmation from the ship's engineers. It is expected that anomalies will be between 1 – 5% of the total dataset. 422 records had 2 or more outliers, and this is within the 5% of the total dataset.

Machine Learning Methods

Isolation forest and OneClass SVM Machine Learning techniques were applied

One Class Support Vector Machine (SVM)

ML model used primarily for anomaly detection. PCA was completed to reduce the dataset from six features to two principal components. This makes it easier to identify patterns in a 2D plot of the data.

This method identified 978 outliers with gamma value of 0.5 and nu 0.05. These represent 5% of the total data. The method was repeated with different values for the gamma and nu parameters. Low gamma values flag fewer anomalies.

The figures below shows the PCA plot of the data with the outliers.

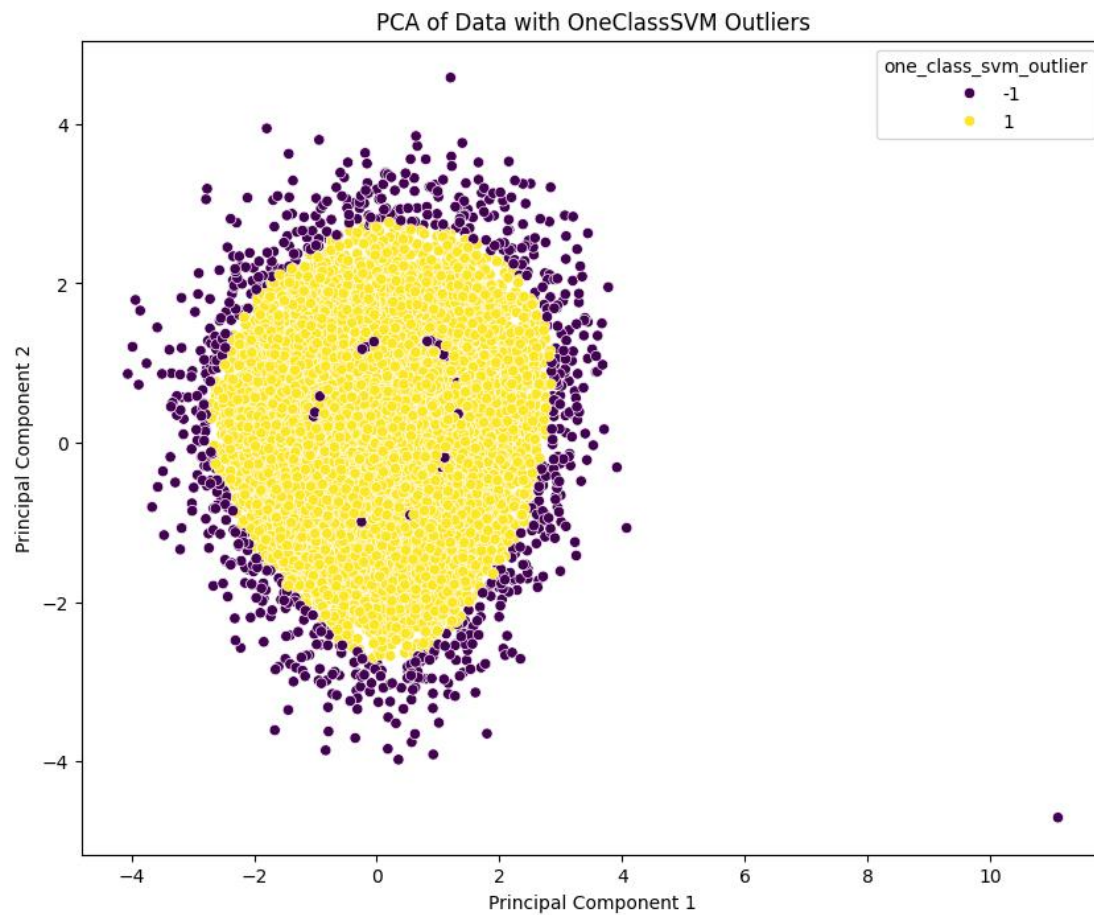


Figure 2: One ClassSVM with Outliers (purple colour)

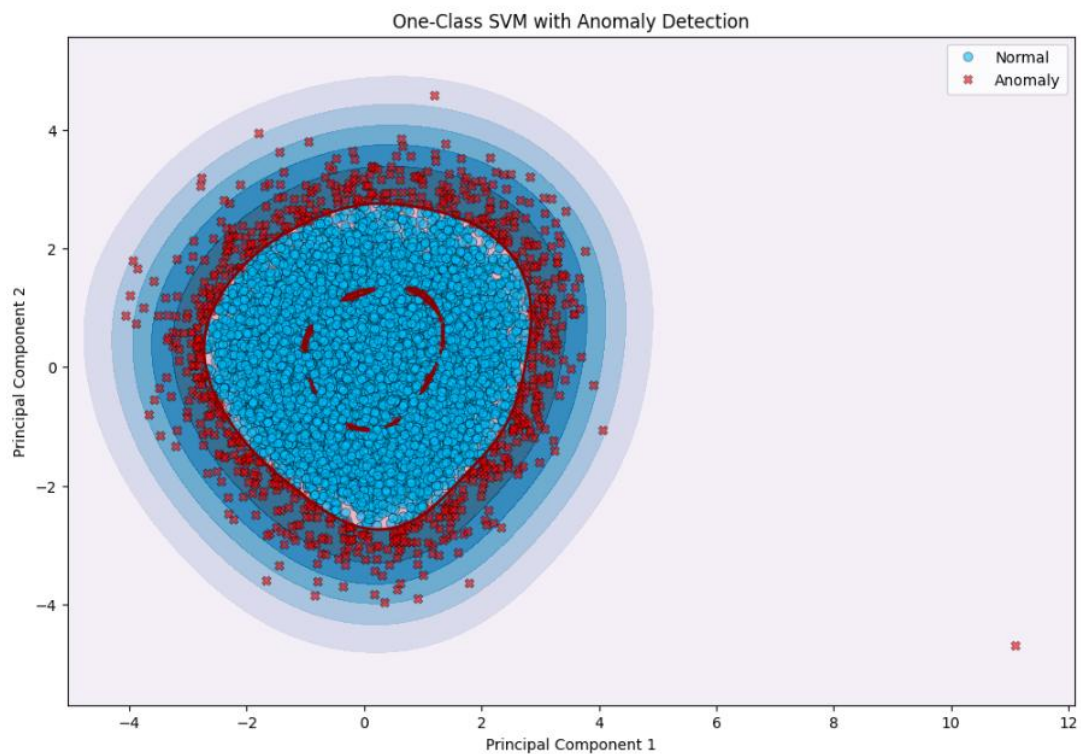


Figure 3: One-Class SVM with Anomaly Detection

Isolation Forest

Isolation Forest was applied with different contamination values. 977 records with outliers were identified, representing 5% of the dataset.

The table below shows the different values obtained from tuning the isolation forest parameters

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Tuning Isolation Forest with different contamination values:  
Contamination = 0.01: Number of outliers = 196 (1.00%)  
Contamination = 0.02: Number of outliers = 391 (2.00%)  
Contamination = 0.03: Number of outliers = 587 (3.00%)  
Contamination = 0.04: Number of outliers = 782 (4.00%)  
Contamination = 0.05: Number of outliers = 977 (5.00%)
```

Table 3: Isolation Forest Outliers

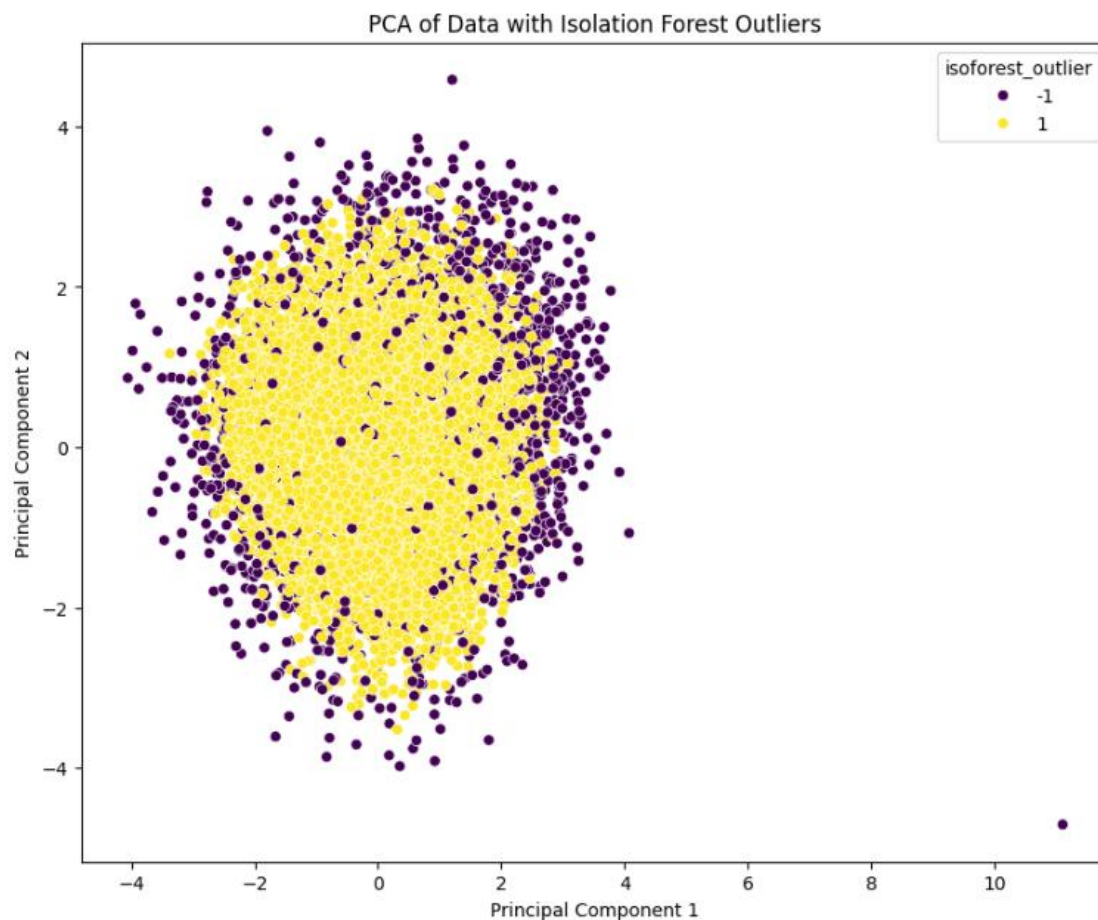


Figure 4: PCA of data with Isolation Forest Outliers

Results

The three methods help to identify outliers by highlighting them. The 2D charts provide visualisation of the outliers and enable the organisation to spot anomalies.

The IQR method identified 422 rows as having 2 or more features classified as anomalies. This is 2.16% and falls within the expected range.

The One-Class SVM method identified 978 (5%) of the rows as having anomalies, whilst the Isolation Forest identified 977 records..

The IQR method was easy to use taking into consideration the business rule that rows with two or more features as anomalies would be treated as anomalies.

The One-Class SVM and Isolation Forest methods helped in identifying outliers and visualizing the anomalies in the data.

Conclusion

This report shows how the organisation can use statistical and machine learning techniques in anomaly detection of ships engines. The analysis, coupled with domain knowledge from stakeholders within the company to verify the outliers, will help the organisation to save money and build trust with customers by detecting anomalies and addressing them promptly.

The company should monitor values of lubrication oil pressure, coolant pressure and fuel pressure as these were more likely to be present in the anomalies detected and may indicate a potentially problematic scenario that requires attention.